

Senior Project Proposal

CENG 4266 Senior Project 2
Computer Engineering Program
University of Houston Clear Lake

MEMORANDUM

{Date}: January 23, 2026

TO: Dr. Luong Nguyen
Instructor

FROM: {Student Name 1}: Zabdiel Montano
{Student Name 2}: Andrew Patino
{Student Name 3}: Randy Rodriguez
{Student Name 4}: Nicholas Boggs

SUBJECT: Senior Project Proposal Submission

Enclosed is our group's design project proposal, LiDAR Scout Rover for Autonomous Environmental Mapping. This proposal is submitted for partial fulfillment of the Senior Project requirement outlining the plan for the project pursuit through the problem formulation with functional requirements, alternative solution generation with electrical and computer engineering approaches, project management and milestones, and task assignments and deliverables. We understand this proposal, in written report as attached and oral presentation upon scheduled, would undergo a rigorous Proposal Review Panel assessment, and we are willing to accept recommendations from the Panel Review and modify and resubmit for final approval.

Senior Project Proposal

{Title}: LiDAR Scout Rover

Submitted by

{Student 1}: Zabdiel Montano
{Student 2}: Andrew Patino
{Student 3}: Randy Rodriguez
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Approved by

Senior Design Advisor:

Luong Nguyen

Name

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{Title}: LIDAR SCOUT ROVER

1. Abstract

The LiDAR Scout Rover project aims to advance the capabilities of autonomous mobile platforms for high-precision environmental mapping and hazard detection. While Semester 1 established a robust hardware foundation using a tracked chassis and a Raspberry Pi 5, the rover currently relies on a wired connection for its LiDAR data pipeline. To address this limitation, Semester 2 focuses on achieving full mobility through wireless data integration and implementing Simultaneous Localization and Mapping (SLAM). The rover integrates advanced sensing technologies, including 4D LiDAR, camera, and gas detection modules, to identify structural hazards and environmental risks such as gas leaks among other things. By providing a safer and more efficient method for exploring inaccessible or dangerous areas, the rover enhances emergency preparedness and offers students valuable experience in ROS 2, embedded systems, and autonomous navigation research. Ultimately, this project demonstrates how compact robotics can provide reliable perception and situational awareness in high-risk scenarios.

2. Introduction

Situational Awareness for Emergency Responders

- Search and rescue teams could deploy the rover ahead of time to generate 3D spatial maps of unfamiliar or hazardous buildings.
- Real-time point cloud data allows for the detection of structural obstacles and safe passage planning without risking personnel.
- Integrated sensors provide critical environmental data, such as identifying obstacles or airborne hazards, before human entry.

3. Problem Statement

Bridging the Gap in Autonomous Perception

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- Many small-scale robotic platforms lack the sensing and processing power required for reliable navigation in unpredictable environments.
- Current prototype limitations restrict the rover's ability to explore untethered.
- There is a critical need to equip mobile robots with sufficient onboard intelligence to interpret point cloud data, avoid obstacles, and operate autonomously without human intervention.

4. Methodology

The methodology for Semester 2 focuses on the transition from a sensing prototype to an autonomous mobile platform capable of real-time SLAM. The project is divided into key technical phases distributed among team members to ensure system integration and validation

1. **Zabdiel Montano – ROS 2 Web UI:**
 - Performs ROS 2 bring-up & TF tree setup.
 - Build the web dashboard.
 - Implements controls: E-stop, waypoint start/stop.
2. **Nicholas Boggs – CAD, 3D Modeling, Chassis Assembly:**
 - Applies wiring standards & interference mitigation for reliability.
 - Designs 3D CAD mounts & brackets for sensors/electronics.
 - Integrates the chassis & tracked platform, aligning drivetrain.
 - Manages power distribution (6S LiPo → 5V/12V buck).
3. **Andrew Patino – System Design/Sensor Communication:**
 - Assemble the circuits and integrate sensors with Pi.
 - Build ROS 2 publishers for camera & gas.
4. **Randy Rodriguez – SLAM/Navigation:**
 - Configures SLAM Toolbox for real-time online mapping.
 - Integrates custom C++ planners/controllers for path following.
 - Develops coverage & route generation
 - Adds hazard-aware slowdown/avoidance for safety.

Each team member will contribute across all phases through group design reviews, integration tasks, and field testing. The final system will be validated in scenarios where drones are less effective, such as narrow hallways, stairwells, and collapsed structures. A block diagram of the rover system will be developed to illustrate subsystems, including mobility, sensing, control, and communication modules. This structured methodology ensures a clear division of labor while emphasizing collaboration toward the shared objective of creating a rover that enhances situational awareness.

5. Resource Requirements

To successfully implement the autonomous navigation phase, the following resources are required:

- **Hardware:**
Tracked chassis with high-torque DC motors, BTS7960 drivers, and 6S LiPo battery system. Unitree 4D LiDAR L1, Waveshare LWIR thermal module, and MQ-series gas

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sensor kit. Raspberry Pi 5 (8GB) with active cooling and high-gain wireless network adapters.

- **Software:**
ROS 2 (Jazzy) runs on a Linux-based environment. POINT-LIO for odometry and RViz2 for visualization.
- **Facilities and Space:**
Access to the University of Houston-Clear Lake Senior Project Lab and campus environments for mobility testing.
- **Funding:**
Budget allocation will be required for hardware components (motors, sensors, batteries, microcontrollers), software tools, and prototyping materials. Additional funding may be necessary for replacement parts and long-term testing.

By securing these resources, the team will ensure the LiDAR Scout Rover evolves into a reliable, autonomous platform for advanced environmental mapping.

6. Schedule

Semester 1 – Core Rover Development

Wk1–2: Researched optimal mobility design and compatible hardware/software.

Wk3–4: Created proposal and finalized budget.

Wk5–6: Ordered major components (LiDAR, sensors, chassis) and configured Raspberry Pi5.

Wk7–8: Installed Ubuntu + ROS 2 dependencies; assembled chassis and wiring schematic.

Wk9–10: Mounted hardware, verified LiDAR operation and stable power.

Wk11–12: Integrated ROS 2 communication and began motor testing.

Wk13–14: Finalized wiring; achieved ROS 2 motor control.

Wk15: Delivered end-of-semester report and presentation.

Semester 2 – Autonomy & Field Testing

Wk16–17: Refined ROS 2 motor control and navigation response. Add gas sensors to the rover with proper coding.

Wk18–19: Linked LiDAR mapping with motion control for path following. Add camera to the rover with proper coding.

Wk20–21: Enhanced web dashboard and Wi-Fi connectivity. Re-organized wiring to be more efficient and compact.

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Wk22–23: Finalize testing for camera + gas sensors; validated data accuracy.

Wk24–25: Re-designed and printed improved chassis/mounts.

Wk26–27: Performed full-system testing and data collection.

Wk28–29: Compiled final report and demo materials.

Wk30: Presented final project and submitted documentation.

7. Budget

While the current expenditure totals \$796.02, additional funding will be required to cover all remaining planned modular components.

8. Contacts

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